Abdominal Transplant

Pre-CA1 education day

I am ranking I'm one of the transplant faculty. this topic is kind of awkward because it has to bridge between sets of knowledge. the first is what do you need to know to do well on the exams as someone who is unlikely to do anesthesia 4 liver transplantation, and the second is what you would need to know to provide really expert care for liver transplant recipients.

Be not afraid

Your attending will help you. Questions are expected.

Your tech knows the defaults for set up.

There is an *exhaustive* protocol.

Some procedures will take longer … that's ok.

Most of our liver transplant recipients *aren't that sick*

You don't need to already know how to use the ROTEM.

It's ok to be nervous on your 1st / 2nd rodeo.

You don't need to be there an hour ahead of time. I go see the patient at T - 15 minutes.

There *almost always* is a pre-evaluation or consult note from anesthesia. Review:

> T&S (antibodies?); ECG (QTc); last EGD (banded in last month?); Hgb, plt, Cr, Na, K, INR, fibrinogen; TTE (PASP).

You don't need a bunch of drips. NE and a tube.

You don't need a pre-procedure arterial line.

You don't need to order blood to the room *before getting to the room.*

A main thing that I want you to take away is that you don't need to be afraid of this. liver transplantation is obviously a major case, and it is one of the bigger cases that does not explicitly require people to have done their cardiac rotation. there are multiple lines of safety to make sure that we do this in a highly reliable way.

The protocol is pretty good

Anesthesiology

Orthotopic Liver Transplantation - Adult

Approval: 1 Aug 2023 by CR King

Surgical Indications & Procedure

- Chronic or decompensated cirrhosis from fatty liver disease (NASH), alcohol use, autoimmune hepatitis, viral hepatitis, rejection of prior graft, portal vein thrombus, iron overload, venous stasis, rare metabolic diseases.
- Biliary disease: PSC, uncommonly PBC, post-transplant biliary ischemia
- Curable liver cancer: within defined criteria, hepatocellular carcinoma, cholangiocarcinoma, oligo-metastatic colorectal cancer.
- Uncommon exceptions like polycystic disease, porto-pulmonary hypertension, and hepato-pulmonary syndrome.

The liver is mobilized through a chevron incision (dissection phase). After the vasculature is isolated, the hepatic and portal veins are transected and liver removed (anhepatic phase). The graft retro-hepatic IVC is end-side attached to the recipient IVC and portal veins anastomosed (reperfusion, piggy-back anastomosis). The hepatic artery and biliary anastomoses are then made (neo-hepatic phase). Rarely, the recipient IVC is transected above and below the liver and the graft sewn to both (bicaval anastomosis). Most OLT are closed primarily. Rarely, a robotic approach is used; see the appendix for the special room setup.

Choice of Anesthesia

General

FAQ and less common topics aka "Why isn't it just like cardiac?"

Recipients and special aspects of end stage liver disease

What is MELD?

The Model for End-Stage Liver Disease was initially developed to predict survival after a TIPS procedure. The MELD score predicts mortality in patients with ESLD and is based on INR, serum bilirubin, serum sodium, and serum creatinine. The MELD (or "lab MELD" to exclude exception points) is widely used to quickly summarize the severity of a patient's liver disease; higher MELDs anticipate more use of vasopressors, surgical blood loss, and difficulty with reperfusion. 80% to 90% of hepatic functional capacity must be lost before overt hepatic failure (synthesis, storage, metabolic, and excretory functions)

OLT indications

Acute liver failure, primary graft non-function

ESLD with biochemical decompensation (MELD > 15)

ESLD with severe, reversible complications

- Hepatopulmonary syndrome, porto-pulmonary hypertension

Cholangiocarcinoma, hepatocellular carcinoma, resectable colorectal cancer

Cystic fibrosis

Hepatic artery thrombosis

Familial amyloid, several metabolic diseases, primary hyperoxaluria

So far as interns you have probably taken care of patients who have had liver transplants in the SICU or who need liver transplants on the medicine service. unfortunately, many of the patients with end-stage liver disease who you have taken care of will simply not be eligible because they have a disease that is not curable. often that as alcohol use disorder, although sometimes it is cancer that is no longer curable. you have probably seen the m e l d score used to discuss the severity of someone's liver disease, and in patients who have primarily a synthetic dysfunction disorder that is still used as a summary. A MELD 15 = net survival benefit, but you will see many patients with other complications of cirrhosis or related diseases who are eligible candidates for transplantation.

OLT indications - emerging and nonstandard

Polycystic liver disease (pain)

TIPS refractory ascites

Allocation

Liver: sickest first

Acute failure: 1a or MELD=40

MELD [2002]: log(bili) + log(INR) + log(Cr)

MELDNa [2016]: MELD + (137 - Na) + M*(137-Na)

MELD3.0 [2023]: + female + albumin + albumin:[Na + log(Cr)]

Exceptions: MMAT - 3

Kidney: Time on dialysis or waitlist

Longevity matching

Exceptions for donors, kids, difficult xmatch

Allocation Circles

Replaced donor service areas

Sequentially offer a donor to recipients in

[1a , 37+ , 33+ , 29+ , 15+] in radii of

[150, 250, 500] nm.

DCD or donor age > 70: just [150, 250, 500].

Peds first within each "bucket"

Order by MELD within "bucket"

Variances for AK, HI, PR

Each Transplant Center has a responsibility to evaluate potential transplant recipients and submit them into the national league organized list. unlike kidney transplant, which is largely based on time, for liver transplant wait list is something of a misnomer, because Acuity is based on expected survival without a transplant.
I mentioned that an m e l d score of greater than 15 is where there is usually a survival benefit, but most patients with liver failure will be substantially higher. we allocate livers in a sickest First fashion mostly based on their laboratory values, although there are many exception point criteria to increase someone's position on the list that you do not need to know at this stage. Sickest first != net benefit, but almost
Meld originally for TIPS survival (small cohorts!). Adopted to replace CP because objective. More and more nonlinear / ML models. Oddly, require validation in non-transplant candidates
donor organs come largely from our local area but can be sourced from up to 500 nautical miles away.

What actually happens

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Room set up:

Induction: Prop+roc+lido

Available: Vaso, insulin, "liver box," zosyn

NE infusion

McGrath, normal sized ETT

Quad lumen CVC

Arterial/CVP pressure bag

A-line: tegaderm, needle, 4x4, prep stick, gloves

RIC: tegaderm, 20g, RIC, prep, gloves, towels

NGT

US, TEE (one is permanently in 225)

One thing that I think stresses people out about their first couple of liver transplants is the idea that they need to be here really early and to get a bunch of stuff set up.

however, because there is a relatively High cancellation rate for transplants when they take the organ out of the donor and find that it is not suitable, we actually don't want you to set up a bunch of things. the expectation is that we will get to the or and help each other.

I personally show up 30 to 15 minutes before a transplant so that I can sit down and talk with them and see if there is anything that needs to be addressed. when I was the fellow and did around 80 of these in a year, I would show up around 30 minutes ahead of time get set up over the course of around 15 to 20 minutes, and go talk with the patient before rolling back. it is uncommon that patience need a pre procedure arterial line.

there are some specifics listed here, but unlike cardiac and lung transplant, our liver transplant recipients almost all go through CPAP or they have an inpatient evaluation by one of the transplant attendings which shows up as a consult note.

these patients will often have millions of dollars of lab work that has seemingly been done but there's actually a relatively short list of things that I would expect you to review the day of service, most of them are in the anesthesia labs summary. the most important one to check usually is if they have developed antibodies on their typing screen.

you also definitely should not expect to set up a million drips. almost everyone will get induced with just a vasopressor and possibly a carrier. the blood ordering is almost completely standardized, and the only thing that trips people up is occasionally bringing platelets to the room that we intended to stay downstairs, unlike

in cardiac surgery.

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NGT

Tech: Arterial/CVP bag, belmont, Quad lumen

US, TEE, twitch view, BIS (permanently in 225)

As I mentioned, the room setup itself is not at all involved and should not take you very long. 98% of the time you need the completely standard set of induction medications norepinephrine set up and then a handful of additional medicines that the pharmacy will get out when you say that you're doing a liver transplant. completely normal intubation equipment and then a couple of supplies for an arterial line and a rapid infusion catheter if you want to do one of those. the tech will set up a lot of standard equipment. you can't go wrong by just asking them to open a quad lumen, and then if you need to double stick to patient and we can just hand you a cordis. there is a fair amount of equipment that lives permanently in 225 so there's no need for you to go hunting for it.

During the case

Dissection: communicate closely, labs hourly

Anhepatic: communicate closely, give steroids, labs when out of ice

Neohepatic: communicate closely

Post-reperfusion: communicate closely, labs as needed, usually hourly

Talk about:

All the time:

Volume goals

Coagulation and platelet goals

Acid-base goals

Na, Ca, K

Anhepatic: everything

Post-reperfusion: extubation

A liver transplant can be divided into four phasers the first phase is when they are chipping the liver out called the dissection. the second phase is when they take that liver out and get ready to put the new one in. the third phase is the first 15 minutes or so after the new one is in. The final phase is the rest of the case where they get the artery and the bile duct hooked up and close the patient. the major expectation during the entire case is that you are going to communicate closely with the attending who will either have no other room or something minor that they don't need to pay attention to. There are a lot of details about what we're doing in transplant, but most of the time if you are talking to me frequently about the volume resuscitation, the coagulation acid base and electrolyte issues, that is going to be most of it. we end up Excavating around half of these patients, and it is often difficult to tell who is going to be ready to extubate until the final hour or so, so unless something has gone very difficult you can just tee everyone up as if they were going to extubate and then be sad about it if they're not ready.

ABA Stuff

4. Gastrointestinal/Hepatic Systems

b. Clinical Science

2) Hepatic Disease

a) Preoperative Laboratory Assessment

b) Anesthesia Choice (Hepatocellular Disease, Ascites, Portal Hypertension)

c) Postoperative Hepatic Dysfunction, Hepatic Failure, Hepatorenal Syndrome

d) Hepatic Transplantation

5. Renal and Urinary Systems/Electrolyte Balance: Clinical Science

a. Renal Disease

4) Anesthetic Management in Renal Transplantation

One way to come at this topic is to ask what do you need to know for the boards. that is probably the easiest question to answer: almost nothing. what you see on this slide are the only topics on the ABA content outline relating to abdominal transplant. you are much more likely to be asked questions about hepatic physiology or end-stage liver disease for non-transplant surgeries. one thing to remember is that many anesthesia residents are at programs that do know or very few liver transplants, and will never see one. Many of the practices in liver transplant anesthesia have never been subjected to rigorous trials, and vary center to Center. which also makes them bad topics for a national standardized exam.

you are obviously responsible for understanding how to do kidney transplants, but there is relatively little that makes a good question about kidney transplants.

ABA Stuff

2. Organ Donors: Pathophysiology and Clinical Management

a. Circulatory

b. Endocrine

c. Fluid and Electrolyte Management

d. Brain Death Criteria

It isn't quite in what we would think about as the same topic, but you are reasonably likely to be asked questions about the management of brain dead or severely brain injured individuals who are organ donor candidates. again, this is the content outline material which is underneath the critical care section of the outline.

Transplant topics

General OLT criteria

MELD, CPT -> waitlist

Neat but low-yield topics

Acute liver failure - intracranial pressure

Transfusion management in ESLD

Temperature management

Anti-rejection meds

Electrolyte / acid management

Reperfusion management

Sodium management

Related topics

TEG / ROTEM

MTP

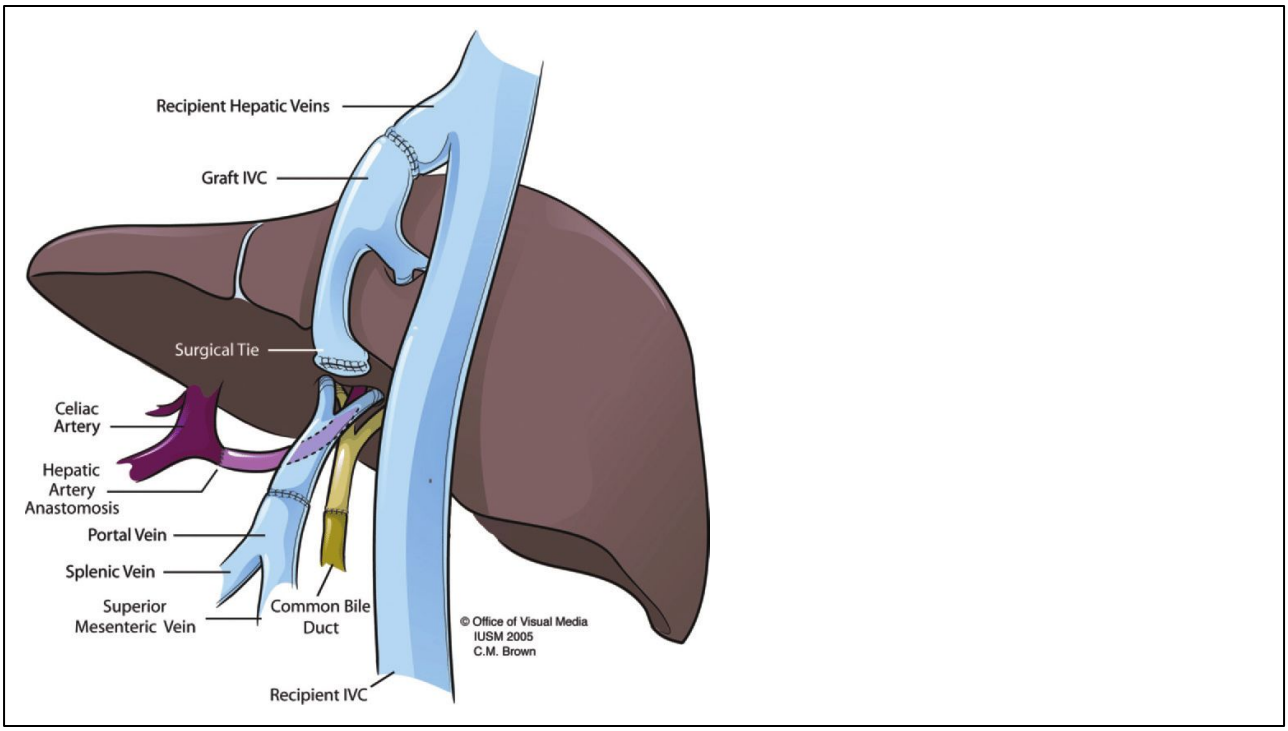
TEE interpretation

ESLD Physiology

"Knocking out the boards" has a nice review of ESLD topics

- Cardiac
- Electrolytes
- Renal function / hepatorenal type 1 vs 2
- Portal hypertension and ascites
- Hepatopulmonary
- Portopulmonary HTN
- Coagulation

So we end up with this division of topics where there are a tiny number of things to know that are both specifically about liver transplantation and fair game for your exam. there are lots of topics that come up in liver transplantation that are on your exam including a fair amount of end-stage liver disease physiology, which are reasonably high yield to review, But are not really specific to transplant. there are also topics that are kind of interesting and important to the actual management of livers, but just very unlikely to show up on your exams


Recipient Hepatic Veins
Graft IVC
Surgical Tie
Celiac Artery
Hepatic Artery Anastomosis
Portal Vein
Splenic Vein
Superior Mesenteric Vein
Common Bile Duct
Recipient IVC
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Kidney transplant

- Poorly justified concerns about platelet function: GA
- Dialysis assessment
- Lab assessment
- Very high rate of CAD
- Immunosuppressants
- Sevo doesn't matter
- Standard ESRD drug considerations
- Roc / vec are *fine.* Sugammadex is fine (maybe slightly slower).
- *Assess for moderate* fluid expansion
- NAC, fenoldopam, mannitol all negative or inconclusive data
- Ephedrine vs others unclear

Dopamine

Increases HR, CO. Mixed vaso-dilation and -constriction.

In *healthy* research participants *increases* renal blood flow (very high quality data)

In *kidney donors* improves post-transplant function (very high quality data)

In *kidney recipients* intraop and postop dopamine does not improve function and likely worsens complications (moderate quality data)

- https://pubmed.ncbi.nlm.nih.gov/8424517/
- https://pubmed.ncbi.nlm.nih.gov/28072739/
- https://pubmed.ncbi.nlm.nih.gov/10617237/
- https://pubmed.ncbi.nlm.nih.gov/20005360/
- https://pubmed.ncbi.nlm.nih.gov/15940046/

In patients with vasodilated or cardiogenic shock, dopamine does not improve or worsens renal function and increases tachyarrhythmias vs norepi (high quality)